Intended and Unintended Effects of Vaping Tax Across Age Groups: Evidence from British Columbia

Ariza Hossain *Mashiat Abonty* *Rain Hert* *Seth Edney*

Abstract

This paper studies how a sharp provincial tax on vaping products changed nicotine use in Canada. In 2020, British Columbia raised its provincial sales tax on vapour products from 7 percent to 20 percent with the stated goal of reducing youth vaping. Using repeated cross sections from the Canadian Tobacco, Alcohol and Drugs Survey and the Canadian Tobacco and Nicotine Survey from 2013 to 2022, we build a harmonised microdata set on vaping, smoking and basic demographics. We compare British Columbia with provinces that did not introduce their own vaping taxes using difference in differences models with province and year fixed effects and an event study for dynamics. For the full population the tax is associated with a small and imprecise fall in past 30 day vaping of about 0.6 percentage points. Among young adults aged 15 to 19, our estimate implies a larger reduction of roughly 1.3 percentage points in recent vaping relative to similar provinces. We find that the tax led to a meaningful rise in cigarette smoking of 0.36 percentage points. The event study indicates that the effect on vaping is strongest in the first year after the tax and then fades. Overall, the results suggest that a one time increase in vaping taxes can decrease youth and young adult vaping in a high cigarette tax environment with an offsetting increase in smoking.

JEL codes: I18, H23, D12, C21.

1 Introduction

Over the past decade, small vaping devices have become a common part of the nicotine landscape. For many adult smokers, e-cigarettes promise a way to move away from conventional cigarettes and their well-documented health harms. At the same time, youth and young adult vaping has grown quickly and often involves flavoured products that don't look like medicine or a quitting aid. This mix of harm reduction for some users and new risks for others sits at the centre of current policy debates. Policy makers want to support adult smokers who switch to vaping, but they also want to prevent a new generation from developing nicotine dependence through vapes rather than cigarettes. A tax on vaping products is one of the few tools that can change choices at the point of purchase, because it raises the relative price of vaping compared with cigarettes. If a higher tax discourages youth vaping but pushes some adults back toward cigarettes, the net health effect is unclear. This makes it important to understand how a large tax on vaping products actually affects smoking, vaping, and overall nicotine use in practice.

Existing research has started to trace out these effects, but important gaps remain. They examine US state taxes and focus on average responses in the whole population. Much of this work uses state-level panel data and difference-in-differences designs to relate changes in taxes to changes in use. The results suggest that vaping taxes reduce e-cigarette use and may raise cigarette smoking, creating a substitution effect. Much less is known about how similar policies work in Canada, where cigarette taxes are already high and where health policy often follows a different path. In such a setting, the substitution for smoking may be more limited. We also know little about how the effects vary by age or about the overall impact on any nicotine use, not just on individual products. This paper addresses these gaps by using a large provincial tax change in Canada to study how different groups adjust their nicotine use when vaping becomes more expensive.

The objective of this paper is to provide new evidence on how a significant increase in a vaping tax shapes smoking and vaping use. We examine British Columbia's January 1st, 2020, increase in its provincial sales tax on vaping products from 7% to 20% as a policy shock and ask three related questions that guide the analysis. First, did this tax reduce the share of people who vape? Second, did the higher vaping tax lead some users to substitute toward cigarettes and, as a result, change overall nicotine use? Third, do these effects differ across age groups, in particular for youth, young adults, and older adults?

To answer these questions, we combine repeated cross sections from two national surveys, the Canadian Tobacco, Alcohol and Drugs Survey and the Canadian Tobacco and Nicotine Survey, covering the years 2013 to 2022 (Statistics Canada, 2023, 2016). We harmonise key measures of

recent vaping, recent smoking, and basic demographics across the two sources. We then compare changes in British Columbia with changes in other provinces that did not introduce their own provincial vaping excise taxes during this period. Our main empirical approach is a difference in differences design with province and year fixed effects, complemented by an event study that traces out year-by-year differences around the tax change. Our identification relies on the assumption that, in the absence of the tax, British Columbia would have followed the same trends in smoking and vaping as the control provinces.

Our results show that the tax led to a fall in vaping of around 0.56 percentage points in BC compared to the control provinces. Isolating for the 15-19 year old range, we found a more sizable effect of 1.3 percentage points. In contrast, we detect an increase in cigarette smoking of around 0.36 percentage points for the whole population. The event study shows that the relative drop in vaping in British Columbia appears immediately after the tax and then diminishes over time.

These results contribute to the policy discussion in several ways. First, they provide, to our knowledge, new quasi experimental evidence on a provincial vaping tax in Canada that follows individual behaviour over a long window. Second, they add an age based view to the growing international literature on vaping taxes and cross product responses. Third, they highlight that a large tax increase can slow the growth of youth and young adult vaping. However, it comes at the cost of a substitution for smoking. Taken together, the findings suggest that the public health effects of vaping taxes may be undesirable, specifically for the youth population.

The rest of the paper is organised as follows. Section 2 reviews the existing literature on e-cigarettes, taxation and substitution between vaping and smoking. Section 3 describes the data and the construction of our main variables. Section 4 presents descriptive evidence on vaping and smoking across age groups and provinces. Section 5 sets out our empirical strategy and reports the regression results. Section 6 concludes with a discussion of the implications for public health and future policy.

2 Literature Review

Over the past decade, a fast growing literature has examined how vaping regulations shape overall nicotine use. This section reviews the main contributions that guide our research question and empirical strategy, with a focus on studies of taxation and substitution between vaping and smoking.

Public health debates about e-cigarettes focus on a tension between harm reduction for adult smokers and the risk of new nicotine use among youth. Early policy work (Brill et al., 2014) argues that

e-cigarettes should not initially face the same excise taxes as cigarettes because existing medical evidence points to much lower toxic exposure and potential benefits for smoking cessation. This work frames optimal e-cigarette taxes in a standard corrective tax framework and stresses that taxes should reflect both external harms and internal costs such as addiction, and that heavy taxes could slow switching away from cigarettes. The analysis is normative rather than empirical, but it sets up the central question in later work. When do e-cigarette taxes reduce harm by discouraging risky use, and when might they make things worse by pushing users back toward cigarette smoking. For our study, this framework highlights that the welfare impact of a vaping tax depends on how it changes both vaping and cigarette use, which motivates our focus on overall nicotine participation rather than on a single product.

A first strand of empirical work studies how taxes on e-cigarettes affect youth tobacco use in the United States. One study exploits US state variation in electronic nicotine delivery system (ENDS) taxes and uses repeated cross sections of high school students (Abouk et al., 2023). The main outcomes are indicators for whether a student vaped, smoked cigarettes, or used both products in the past 30 days. The key independent variables are state level ENDS tax measures and their interaction with post policy periods, while important controls include individual demographics, state cigarette taxes and other tobacco policies, together with state and year fixed effects that capture broader trends. The identification strategy is a difference in differences design that compares changes in outcomes between states that adopt ENDS taxes and those that do not, under the assumption of parallel pre policy trends. The study uses several years of pre policy data to check for trend differences and reports robustness checks that support, but cannot fully prove, this assumption. The findings show that higher ENDS taxes reduce youth vaping but are also associated with higher youth cigarette smoking, which points to substitution between the two products rather than a simple reduction in nicotine use. This provides a useful benchmark for our analysis by showing that taxes can shift behaviour across products, and it underscores the need to track both vaping and smoking outcomes. At the same time, the focus on US high school students and on a particular mix of state policies leaves open whether similar substitution patterns arise in a Canadian setting with different cigarette taxes and regulations and a broader age range.

Young adult responses to vaping and cigarette taxes are studied using US data on people aged 18 to 25, a group that is close in age to many of the vapers in our Canadian data (Friedman and Pesko, 2022). In this work, the main outcomes are indicators for daily vaping and recent cigarette smoking. The key independent variables are state level ENDS and cigarette tax rates, and the models include individual controls such as age, gender and education, as well as state and year fixed effects and indicators for other tobacco regulations. The identification strategy again relies on a difference in differences framework that compares changes across states as taxes move,

with event study graphs used to assess the plausibility of parallel pre treatment trends. The main finding is that higher ENDS taxes reduce daily vaping among young adults but are accompanied by higher cigarette smoking, while higher cigarette taxes have the opposite pattern. In other words, young adults tend to move between products rather than out of nicotine use in response to relative price changes. For our purposes, this study shows that cross product substitution among young adults is empirically important and that tax policy should be evaluated using both vaping and smoking outcomes. It also suggests that age specific estimates can reveal different responses, which motivates our choice to estimate separate effects for young adults and older adults in Canada. At the same time, the focus on US state tax variation and on a relatively narrow age band leaves open how these patterns generalise to a setting with higher cigarette taxes, different regulatory history and a wider age distribution.

A related literature examines how tax policies affect adult smoking and quitting more broadly. One influential study focuses on the first large state level e-cigarette tax in Minnesota, where the state applied a very high ad valorem tax to e-cigarette products (Saffer et al., 2020). The main outcomes are adult cigarette smoking and measures of quit success, while the key independent variable is an indicator for Minnesota in the period after the tax increase. The authors compare adults in Minnesota with adults in other US states before and after the tax using a difference in differences design with state and year fixed effects and controls for cigarette taxes and other tobacco policies. The identifying assumption is that, absent the tax, Minnesota would have followed similar trends in smoking and quitting as the control states, and the paper presents robustness checks with alternative control groups to support this claim. The findings show that the tax reduced adult e-cigarette use but was associated with higher cigarette smoking and lower quit success among adults, which suggests that in this setting the tax may have slowed progress in reducing smoking. This study is informative for our work because it demonstrates that the substitution patterns seen among youth and young adults can also appear among adults and that a vaping tax can affect quitting behaviour. It also shows how a difference in differences design can be used to study a single prominent state policy. However, the Minnesota tax took place in an environment with lower cigarette taxes and a different regulatory context than Canada, so it remains an open question whether similar adult responses occur when cigarette taxes are already high and when only vaping taxes change.

Additional evidence comes from work that uses multi state tax variation among adults to estimate the joint effects of traditional cigarette and e-cigarette tax rates (Pesko et al., 2020). In this research, the main outcomes are adult smoking, vaping and dual use, and the key independent variables are cigarette and e-cigarette tax rates at the state level. The empirical strategy is a panel regression with state and year fixed effects and controls for other tobacco regulations and economic conditions, which aims to separate the effects of each tax while holding other factors constant. The

identification again rests on parallel trends in the absence of policy changes. The study finds that higher e-cigarette taxes are associated with lower vaping but higher smoking and dual use, while higher cigarette taxes reduce smoking and increase vaping, consistent with economic substitution between products. The quantitative elasticities are modest in size but indicate that cross product responses are meaningful in aggregate. For our paper, this work provides a benchmark by showing that adult substitution is present at the national level and by suggesting plausible magnitudes for policy effects. At the same time, the multi state US setting, with its particular mix of taxes and regulations, differs from the Canadian provincial framework, and the study does not examine how effects vary across age groups, which is a key focus of our analysis.

Beyond taxes, several papers examine how other regulations shape vaping and smoking, especially in Canada. One study uses repeated cross sectional surveys of 16 to 19 year olds in Canada, England and the United States from 2017 to 2019 to study youth responses to e-cigarette marketing regulations and flavour restrictions (Hammond et al., 2020). The main outcomes are indicators of current vaping and reported exposure to e-cigarette marketing, and the key policy variables capture differences in rules on advertising, point of sale displays and flavours across jurisdictions. The models use logistic regressions with controls for demographics and tobacco use, and include province or country and year indicators to capture broad differences in policy environments. The cross sectional design limits the ability to make strong causal claims, but the comparisons show that youth who report more marketing exposure have higher odds of vaping and that jurisdictions with stricter marketing and flavour rules experienced smaller increases in youth vaping over time. This study is an important Canadian contribution because it shows that non tax regulations can materially influence youth vaping patterns. Our work complements this by focusing on a fiscal intervention rather than on marketing rules and by including both youth and adults in a single framework, which allows us to examine how a tax based policy operates alongside the types of non tax regulations studied in this paper.

There is less empirical work on Canadian regulatory changes that directly alter product characteristics. One recent study analyses scanner data from Canadian grocery and convenience stores to study responses to the federal nicotine cap that limits most e-liquids to 20 milligrams of nicotine per millilitre (Irvine and Hampsher-Monk, 2022). The main outcomes are sales volumes of different e-liquid products, and the key independent variable is an indicator for the period after the nicotine cap, interacted with whether products comply with the new limit. The analysis uses trends graphs and difference in differences style comparisons across provinces that adopted strong provincial rules before the federal limit and those that did not, with controls for product characteristics and time effects. The findings show a clear shift from higher nicotine pods to products that comply with the cap, and suggest that some users respond by changing the volume they consume

or by switching toward refillable or open systems that deliver nicotine at lower cost per milligram. The study also estimates a substantial loss in consumer surplus for adult e-cigarette users, indicating that the regulation raises the effective price of nicotine. This work is highly relevant to our analysis because it shows how Canadian consumers adjust when the effective price of nicotine rises and how substitution can occur across device types and retailers. Our survey based approach complements this by providing individual level evidence on how a change in vaping taxes affects both vaping and smoking participation, rather than only sales of particular products.

The empirical tax and regulation studies above sit on top of a growing methodological literature on policy evaluation with aggregate and micro data. One influential contribution reviews synthetic control methods and clarifies the settings where they provide credible counterfactuals (Abadie, 2021). The main idea is to construct a weighted combination of control units that closely matches the treated unit before the policy change, and then to compare post treatment outcomes between the treated unit and this synthetic control. Synthetic control methods work best when there is one treated unit, many potential controls and a long pre treatment period, and when interference across units is limited. This methodological work is relevant for our setting because we have one treated province and several potential controls and a relatively long pre treatment window. Although our main analysis uses difference in differences rather than synthetic control, the insights from this literature inform our choice of comparison provinces and our emphasis on examining pre treatment trends and robustness to alternative control sets.

Taken together, the existing literature delivers several clear messages and leaves some important gaps. First, most studies find that e-cigarettes and cigarettes are economic substitutes. Higher e-cigarette taxes or stricter e-cigarette regulations tend to reduce vaping but also increase smoking, both for youth and for adults. Second, the magnitudes are meaningful but not huge. Tax elasticities are often in the range of 0 to 0.2 in absolute value for participation, and policy induced changes in smoking are on the order of one percentage point in many settings. Third, the balance between intended and unintended effects is highly context specific. It depends on baseline smoking and vaping rates, on the design of the tax, on which complementary policies are in place, and on how youth and adults differ in their responses. These patterns suggest that it is important to study e-cigarette taxes in a range of institutional environments and to pay attention both to cross product substitution and to overall nicotine use.

At the same time, there are several gaps that motivate our study. Most of the causal tax evidence comes from the United States. There is little work on e-cigarette taxes in Canada, where the policy mix differs and where provincial and federal rules interact. Existing Canadian work focuses mainly on youth marketing regulations and nicotine limits and uses either aggregate sales data or

repeated cross sectional youth surveys rather than nationally representative adult microdata that observe both vaping and smoking. Very few papers study how the effects of e-cigarette taxes vary across the age distribution using harmonised survey data that contain both products and allow for measures of overall nicotine participation. Finally, there is limited evidence on how a single provincial tax, like the British Columbia 20 percent vaping tax, affects both vaping and smoking relative to the rest of the country when federal policies and other provincial rules are held constant.

Our paper speaks to these gaps in three ways. First, we bring together the Canadian Tobacco, Alcohol and Drugs Survey and the Canadian Tobacco and Nicotine Survey to study vaping and smoking in a consistent way from 2013 to 2022, using harmonised measures of recent use for both products. Second, we focus on the British Columbia tax and estimate its impact using a difference in differences framework with province and year fixed effects, which parallels the identification strategies in the US tax literature but in a new institutional setting with higher cigarette taxes and a distinct regulatory history. Third, we pay special attention to age heterogeneity and to overall nicotine use, asking whether the tax primarily affects youth and young adults, and whether any reductions in vaping are offset by higher smoking or by changes in the share of people who use either product. In this way, our study connects the international literature on the intended and unintended effects of e-cigarette taxes with the specific Canadian experience and with the policy question of how to design taxes that improve public health without creating large harms for harm reduction.

3 Data

This study uses microdata from two national surveillance programs run by Statistics Canada (Statistics Canada, 2023, 2016). The first is the Canadian Tobacco and Nicotine Survey (CTNS), conducted from 2019 to 2022. The second is its predecessor, the Canadian Tobacco, Alcohol and Drugs Survey (CTADS), carried out in 2013, 2015 and 2017. Both surveys are designed to monitor the use of tobacco, nicotine, alcohol and other drugs among people aged 15 years and older who live in the ten Canadian provinces. Each survey uses a probability based, stratified sampling design. Dwellings are selected at random from a national address database. One household member is then sampled and invited to complete the interview. In CTADS interviews are conducted by telephone. In CTNS the main mode is a self administered web questionnaire, with telephone follow up for non response. Different people are interviewed in each year, so the combined file is a set of repeated cross sections rather than a panel of individuals. Each survey year is intended to be representative of the household population in the ten provinces once the survey design is taken into account.

The original survey questionnaires differ in their structure and in the range of topics they cover. At the same time they share a stable core of questions on cigarette smoking, vaping, and basic demographics. We use these common questions to harmonise CTADS and CTNS into a single analysis file. Our main vaping outcome is a binary indicator that equals one if the respondent reports vaping any product in the past 30 days. In the microdata this variable is stored as `VAP_10R`. Our main smoking outcome is a binary indicator for smoking in the past 30 days, recorded as `DV_SSB`. These two indicators capture recent use rather than lifetime or experimental use and are available in a comparable way across survey cycles. We construct age categories using the `AGEGROUP` variable, which places respondents into seven bands. These bands are 15 to 19, 20 to 24, 25 to 34, 35 to 44, 45 to 54, 55 to 64, and 65 years or older. Province of residence is given by `PROV_C`. Each record is linked to a survey year, which we treat as the time dimension in the main regressions.

The policy of interest is the increase in the provincial sales tax on vaping products in British Columbia. In November 2019 the provincial government announced that the tax rate on vapour products would rise from 7% to 20%. The new rate took effect on January 1, 2020. The CTNS field periods line up closely with this timing. The 2019 CTNS was in the field from late October to mid December 2019. It therefore reflects behaviour under the old tax regime and spans the period around the tax announcement. The 2020 CTNS was collected from early December 2020 to mid January 2021, well after the higher tax came into force. Earlier CTADS waves provide a longer pre period with no provincial vaping tax in place. In our empirical work we treat British Columbia as the treated province. Survey years from 2020 onward are coded as post treatment. At each survey wave, the control group consists of provinces that did not introduce their own specific excise taxes on vaping products during the sample period. This design focuses on the contrast between British Columbia and other provinces that did not change their vaping tax treatment, while the inclusion of survey year fixed effects later in the analysis helps to absorb common shocks over time, including those related to the COVID-19 period.

The combined data set contains roughly eighty five thousand individual level observations once all survey years are pooled and basic cleaning is complete. We restrict the sample to respondents aged 15 years or older and drop records with missing values for the main outcomes or key covariates. The final analysis file includes information on vaping, smoking, age group, province, survey year and a small set of other demographic controls that are used as covariates in the regressions. These variables form the basis for both the descriptive statistics and the regression analysis later in the paper.

Both CTNS and CTADS provide survey weights that can be used to recover population level estimates. In this paper we estimate our main regressions without applying these weights. Under standard assumptions, unweighted linear models can recover consistent average effects when the main factors that drive sampling and response are included directly as control variables. In our case we control for province, age group and survey year, which capture important features of the survey design and geography. Working with unweighted models also improves statistical precision. At the same time this choice means that our estimates describe the pooled survey sample rather than providing exact population averages. We view this as a reasonable tradeoff but note it as a limitation.

Several other data limitations should be kept in mind. All measures of vaping and smoking are based on self reports. They may be affected by recall error or by social desirability concerns, especially for younger respondents. Harmonising CTADS and CTNS requires matching variables across survey cycles and aligning response categories. Small differences in wording or in the structure of response options can introduce some measurement noise even when the underlying concept is the same. The shift from interviewer administered telephone surveys in CTADS to primarily web based questionnaires in CTNS may also influence how people report sensitive behaviours, although the use of survey year fixed effects helps to capture broad changes over time. The public use files do not contain exact interview dates. This prevents us from examining very short run responses around the announcement and implementation of the British Columbia tax and leads us to define treatment at the survey year level. Finally, the surveys exclude some parts of the Canadian population. People living in institutions, people living on reserves and full time members of the Canadian Armed Forces are not covered. Our findings therefore apply to the household population of the ten provinces and may not generalise to these excluded groups.

4 Descriptive Analysis

This section builds on the data described above and gives a more comprehensive view of how Canadians vape and smoke. We start by looking at simple patterns in the raw data. These patterns help us see who uses nicotine, where use is more common, and how these behaviours change over time. The goal here is not to estimate causal effects. Instead, we introduce the main facts that shape our later results and that motivate the use of a difference in differences framework. The figures in this section trace out a story about how vaping and smoking fit into the life cycle, how British Columbia compares with other provinces, and why it is reasonable to compare it to a specific group of controls in the empirical analysis.

We start with the life cycle of nicotine use. Compared to vaping, which is often considered a youth phenomenon, cigarette smoking is less dependent on the age of an individual. The first figure confirms this hypothesis and helps us analyze the extent to which particular age groups are susceptible to a policy change regarding vaping.

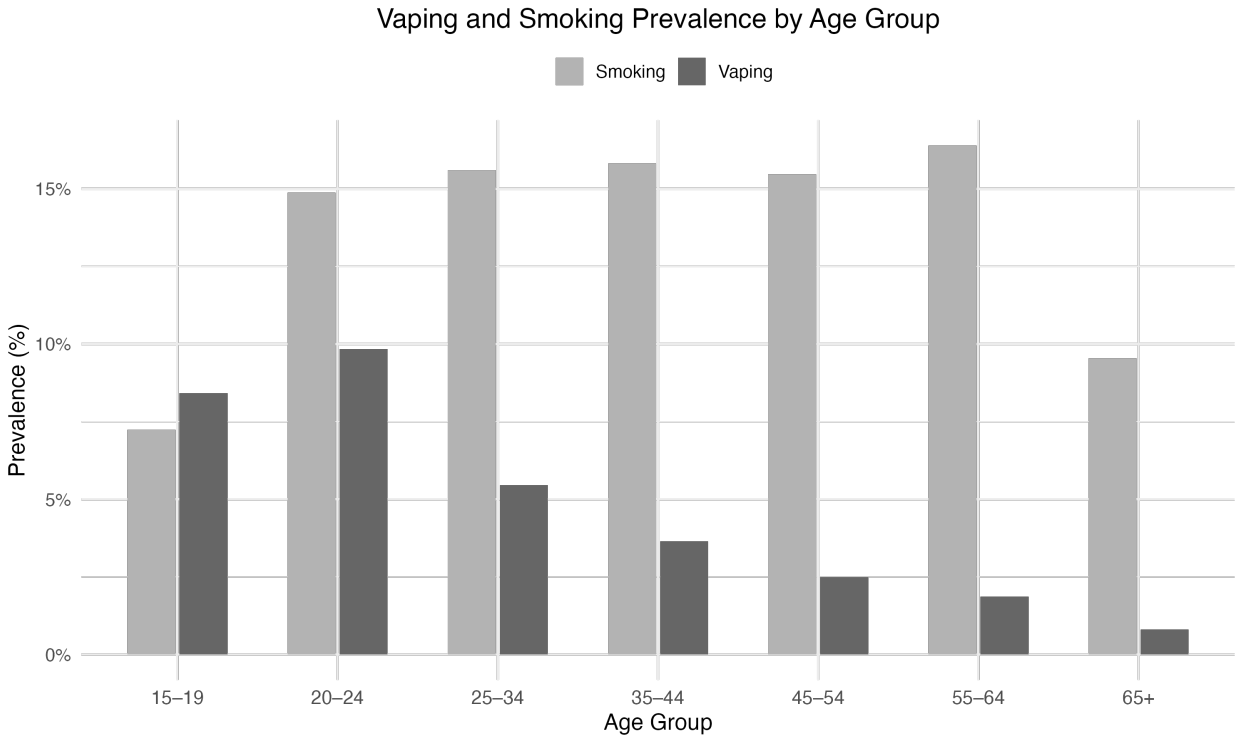

Figure 1: Vaping and smoking prevalence by age group.

Figure 1 shows the share of people who vape and the share who smoke in each age group. Vaping is concentrated among young people. It rises sharply from the late teens into the early twenties, peaks in the group aged twenty to twenty four, and then falls quickly with age. By the time we reach the oldest age group, vaping is rare. Smoking follows a different trend. It rises through the twenties and thirties, stays relatively high through midlife, and only declines at older ages.

This pattern aids our understanding of the differential effect of the vaping tax in British Columbia on different age groups. For many teenagers and young adults, vaping is often their main exposure to nicotine. For older adults, vaping, if it occurs at all, sits on top of an already established history of cigarette use. A tax that raises the price of vaping is therefore likely to affect people at very different points in their nicotine careers. The age profile in Figure 1 suggests that any tax effect will be strongest among people in their late teens and twenties and that we should pay particular attention to how their behaviour changes. It also hints that, for older adults, the main question is

not whether they start using nicotine but whether they adjust the mix between vaping and smoking. To complement this discussion, Figures 10 and 11 in the appendix plot age-specific trends in vaping and smoking in British Columbia between 2013 and 2022. These visualizations deepen our understanding of how nicotine use has shifted over time and allow us to assess whether the tax had especially pronounced effects among younger cohorts, or whether the observed changes reflect broader underlying factors rather than policy alone.

To understand how vaping fits into the broader landscape of nicotine use, we look at the composition of users. Figure 2 categorizes users into three groups: vape only, smoke only, and dual users.

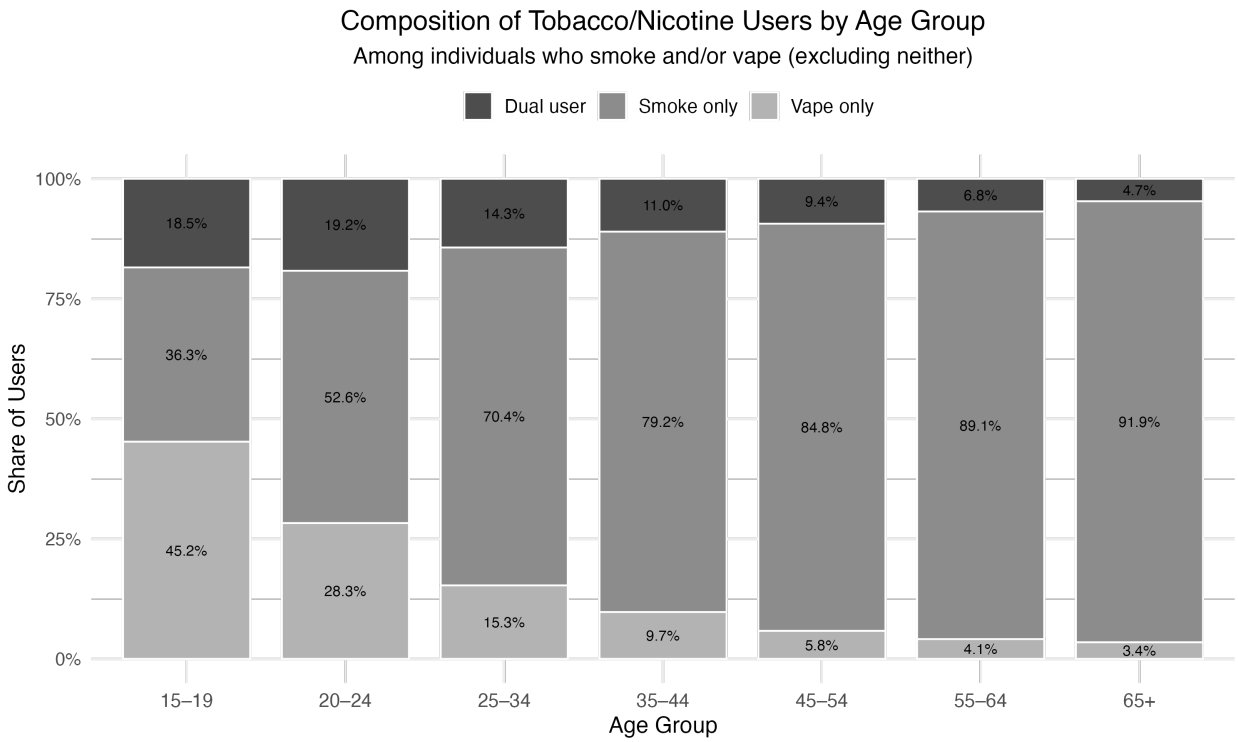

Figure 2: Composition of tobacco and nicotine users by age group.

Figure 2 shows that the mix of nicotine usage varies with age. Among users aged fifteen to nineteen, almost half are vape only. A sizeable group smoke only, and a smaller but important group are dual users who both vape and smoke. Dual use remains common in the early twenties and then becomes much less frequent as age increases. By middle age, most users are smoke only and very few are vape only. In the oldest age groups, nicotine use is dominated by cigarette smoking, with vaping playing only a minor role.

This composition is important for policy. Dual users are the group most able to switch between

products when relative prices change, because they already have experience with both. The fact that dual use is concentrated among youth and young adults suggests that these age groups are more susceptible to substitution towards cigarettes due to an increase in the price of vaping products. For older adults, where smoke-only use dominates, there is less evidence of routine switching between products in the descriptive data. This is one reason why, in the empirical work below, we estimate age-specific effects of the tax and consider outcomes that combine vaping and smoking into a broader measure of any nicotine use.

We now move from age patterns to differences across provinces. Figure 3 shows the share of respondents who vaped in the past thirty days before 2020, averaged across the pre policy years.

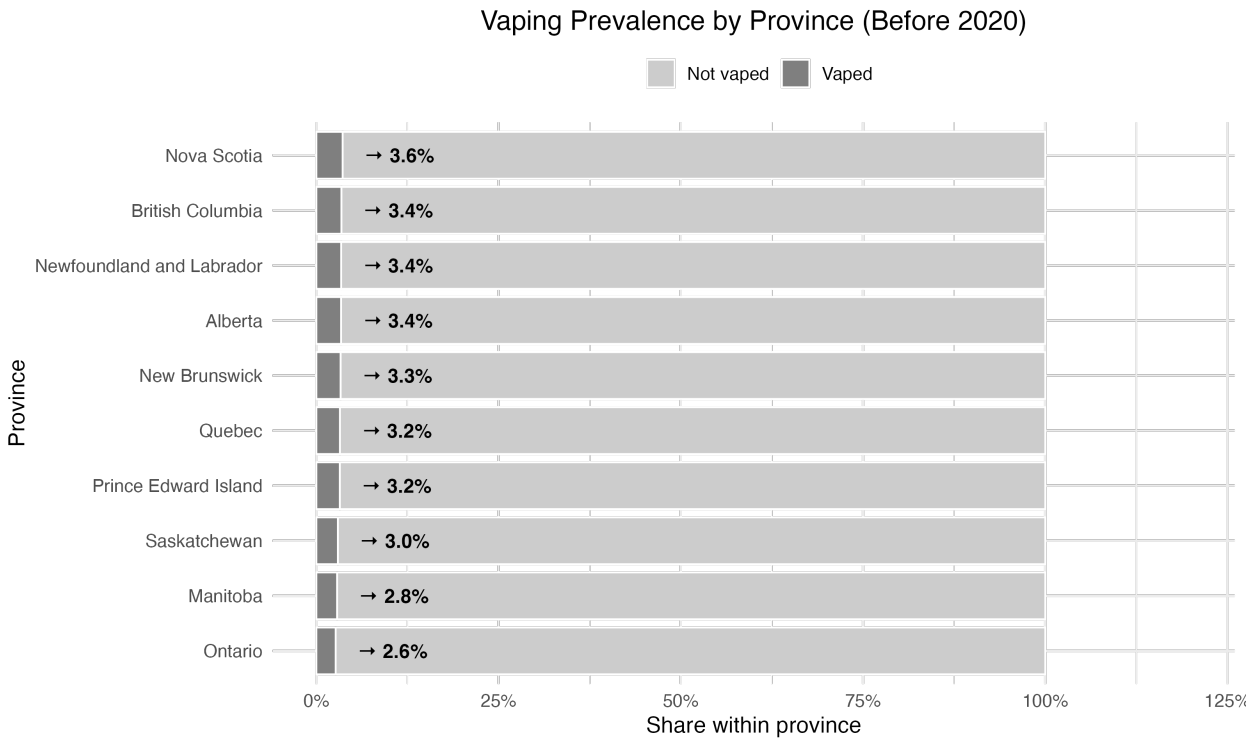

Figure 3: Vaping prevalence by province before 2020.

In the pre tax period, British Columbia stands near the high end of the distribution. Its rate is close to that of Nova Scotia and Newfoundland and Labrador and clearly higher than the rate in Ontario. This tells us that British Columbia begins the period as a province with relatively common vaping. Therefore, the tax arrives in a setting where vaping is already well established, rather than in a province that is trying to prevent vaping from taking off in the first place. This initial point matters to observe the dynamic changes post taxation.

Figure 4 repeats the exercise for the period from 2020 onward, after the British Columbia tax

comes into effect.

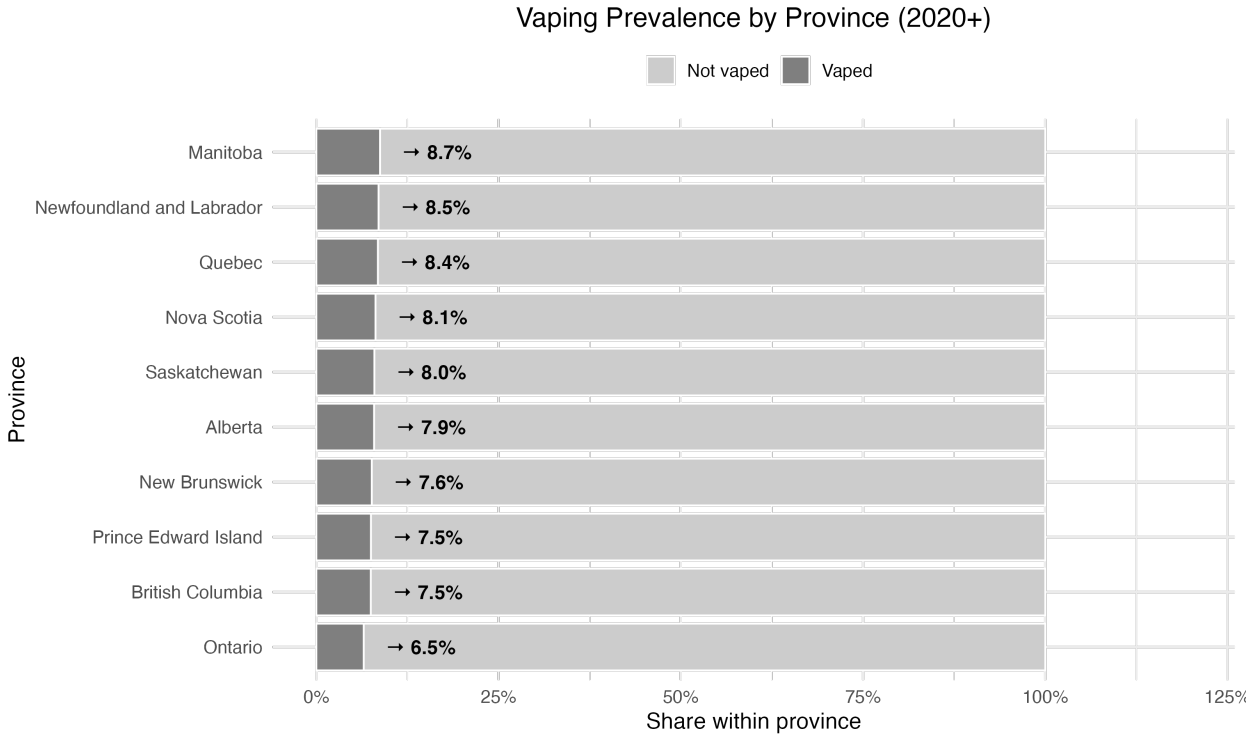

Figure 4: Vaping prevalence by province from 2020 onward.

In the post-2020 period, vaping rates rise everywhere, reflecting the rapid expansion of these products during this time. At the same time, the relative positions of provinces shift: Manitoba, Newfoundland and Labrador, and Quebec move to the top of the distribution, while British Columbia falls into the lower half. Its vaping rate continues to grow, but at a slower pace than in most other provinces. On its own, this pattern does not prove a tax effect. Other provincial policy changes or broader environmental shifts could also be at play. Still, the change in ranking is consistent with the idea that British Columbia transitions from being a relatively high-vaping province to a relatively low one after the tax. Figure 13 in the appendix, which plots overall smoking and vaping trends across all provinces over time, reinforces this context by showing that vaping generally does not decline after 2020 in most provinces.

We next focus directly on British Columbia and trace how vaping and smoking evolved over time. Figure 5 plots both behaviours in British Columbia from 2013 to 2022.

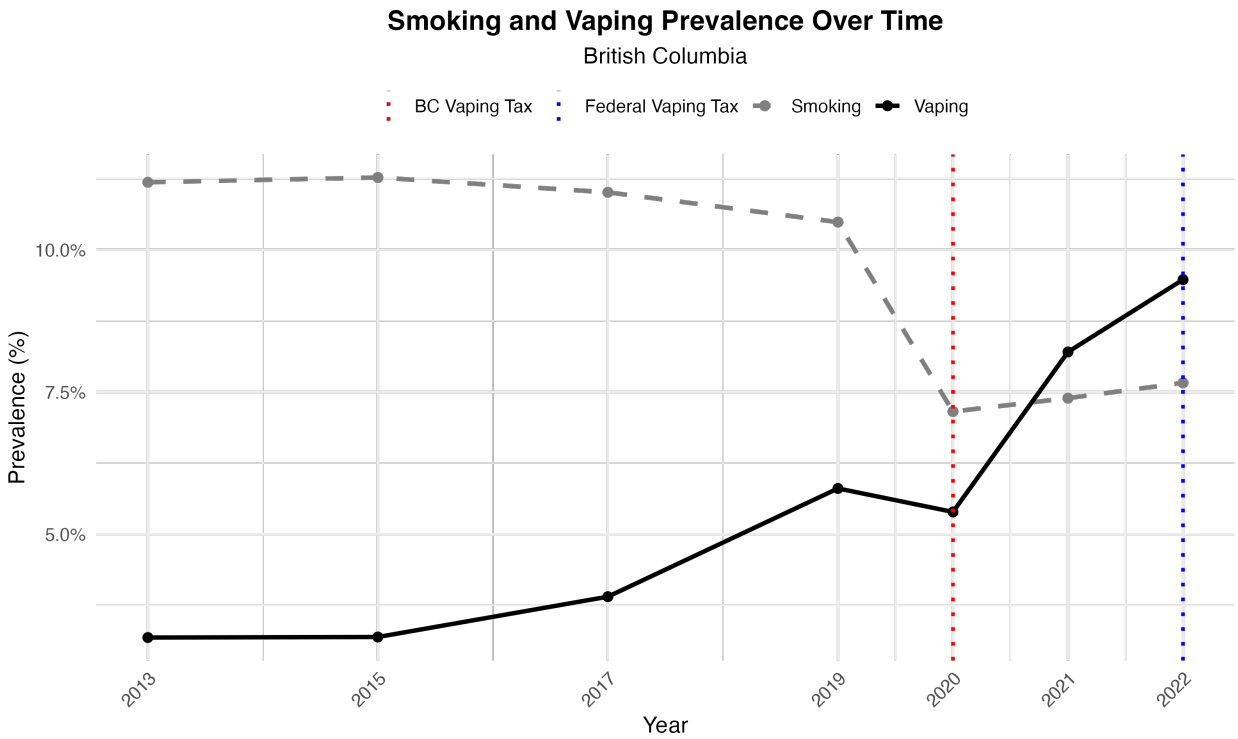

Figure 5: Smoking and vaping prevalence over time in British Columbia.

In Figure 5, smoking declines slowly from 2013 to 2019. Between 2019 and 2020 it falls more sharply, then stabilises and even edges up slightly in the later years. Vaping follows the opposite trajectory. It rises steadily up to 2019, dips modestly in 2020, and then increases again in 2021 and 2022. Vertical lines mark the key events. The provincial tax is in effect at the start of 2020 and the federal tax comes later in 2022. The timing lines up with the drop in smoking and the short pause in vaping growth. At the same time, 2020 is also when the COVID begins, so behaviour may have changed for reasons that are not directly related to the tax. Without a comparison group, it is difficult to separate the tax effect from these broader shocks. The time series for British Columbia alone is therefore suggestive rather than conclusive and points toward the need for a design that compares British Columbia with other provinces.

To place British Columbia in a national context, we compare its vaping rate to the Canadian average. Figure 6 shows vaping prevalence in British Columbia and in Canada as a whole.

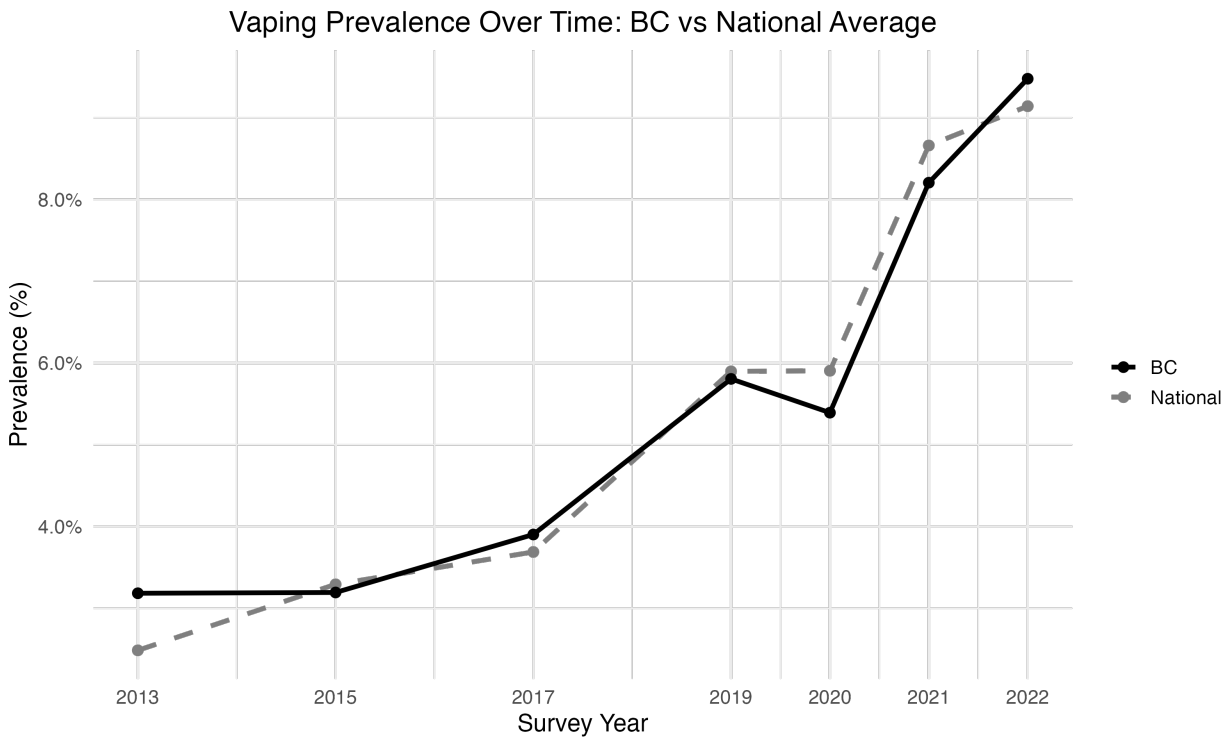

Figure 6: Vaping prevalence in British Columbia and the national average.

Before 2020, the British Columbia line and the national average in Figure 6 move closely together. Both rise from low levels as vaping spreads through the population, and there is little sign of a systematic gap between the province and the rest of the country. This suggests that, in the absence of a province specific tax, British Columbia might have continued to track national trends in vaping. After 2020, the lines separate slightly. The national average continues to rise, while British Columbia sits a little below it. The gap is not large, but it is persistent across the later years. Taken together with the provincial rankings above, this supports the idea that the tax may have slowed vaping growth in British Columbia relative to the rest of Canada, though it does not yet isolate the tax effect from other differences across provinces.

Finally, we compare British Columbia directly with a control group of provinces that did not introduce their own vaping tax during our study period. Figure 7 shows smoking prevalence for British Columbia and the control provinces over time.

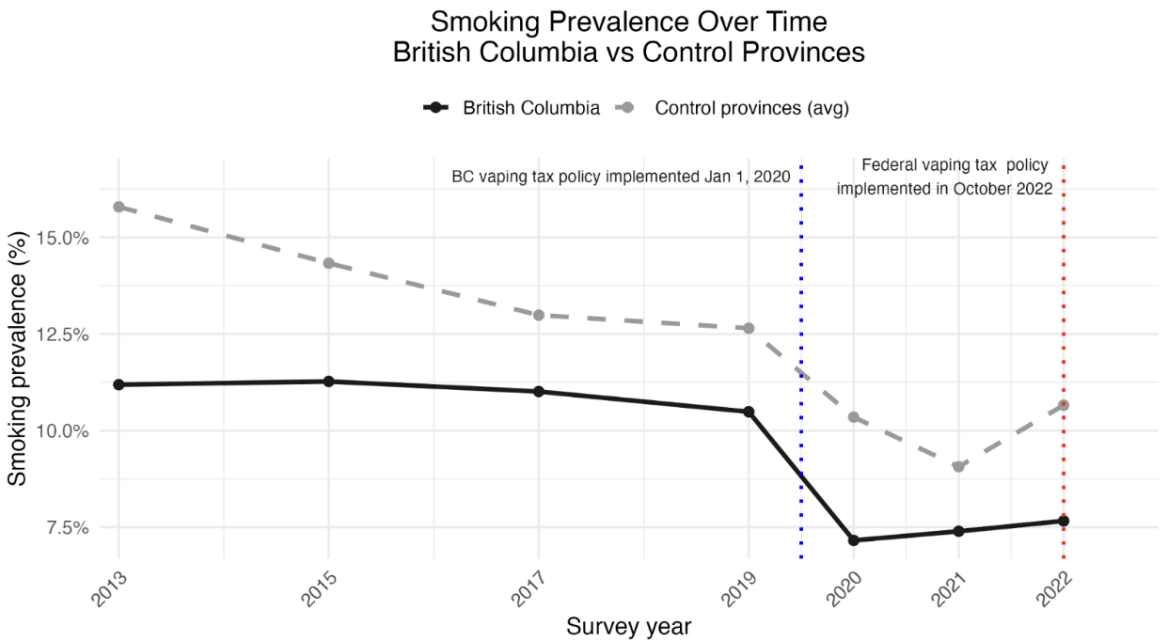

Figure 7: Smoking prevalence in British Columbia and control provinces.

For smoking, both British Columbia and the control provinces show similar downward trends before 2020. The levels differ, but the slopes are close, which supports the idea of parallel trends that underpins the difference in differences approach. Around 2020 both series fall, but the decline is larger in British Columbia. This suggests the possibility that smoking responded more strongly to changes in the local environment, including the introduction of the vaping tax and other contemporaneous factors.

We then examine vaping in the same way. Figure 8 plots vaping prevalence in British Columbia and in the control provinces.

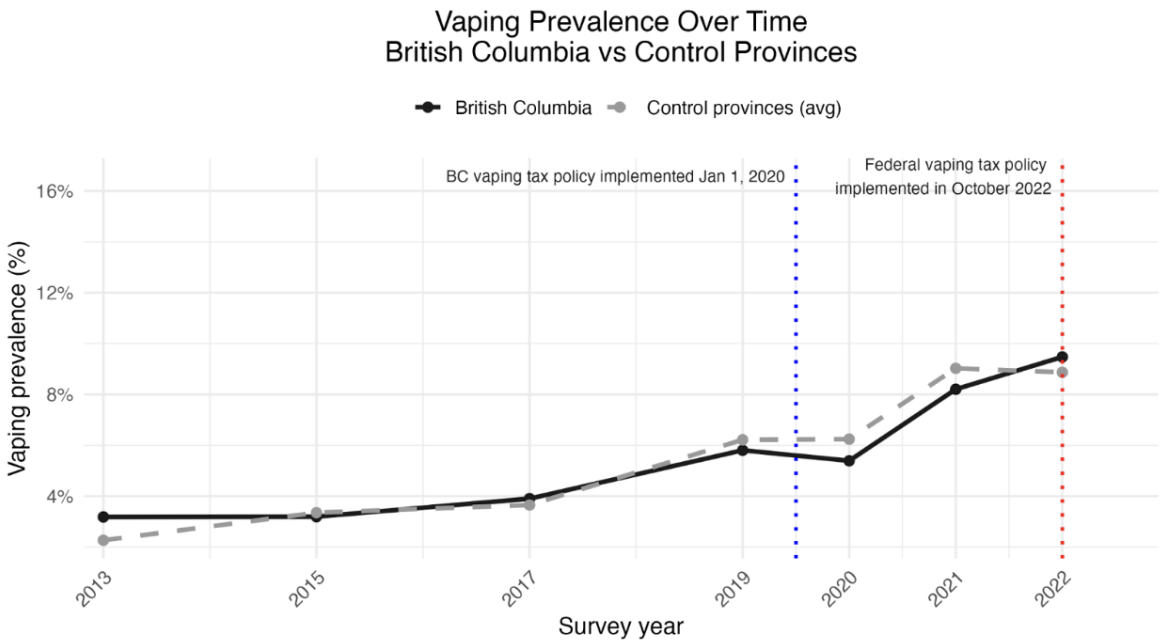

Figure 8: Vaping prevalence in British Columbia and control provinces.

Before 2020, the trends in British Columbia and the control provinces move together in Figure 8, again with only small level differences. This supports the parallel trends assumption for vaping as well. After 2020, vaping continues to rise in both series, but British Columbia stays just below the control group. The gap is modest but persistent across the later years. This small and sustained wedge is consistent with the descriptive and cross sectional patterns we observed earlier, where British Columbia moves from being a relatively high vaping province to a relatively lower one after the tax.

Across all of these figures, a coherent pattern emerges. Vaping is strongly concentrated among younger people, and many young users combine vaping with smoking. British Columbia begins as a relatively high vaping province and then becomes a relatively low vaping province after the introduction of the tax. Its vaping rate grows more slowly than in comparable provinces, while smoking falls sharply at the same time. These patterns are suggestive, but they cannot separate the effect of the tax from other changes happening in the same years. To do that, we now turn to a difference in differences framework. This approach allows us to compare British Columbia with similar provinces while controlling for broader trends over time and unobserved differences across regions, and to quantify the tax effects and their age specific heterogeneity more precisely.

5 Empirical Strategy & Findings

In this section we move from descriptive patterns to a regression framework that aims to isolate the impact of the British Columbia vaping tax. We focus on three main outcomes: recent vaping, current smoking, and an indicator for any nicotine use that equals one for individuals who either vape or smoke. The dependent variables are therefore all binary indicators. Throughout, we estimate linear probability models and cluster standard errors at the province level so that statistical inference allows for arbitrary correlation in outcomes within provinces over time.

We begin with a simple difference in differences model that compares British Columbia to a set of control provinces before and after 2020. For individual i in province p and year t we estimate

$$Y_{ipt} = \alpha + \beta\,\text{Treated}_p + \gamma\text{Post}_t + \delta\,(\text{Treated}_p \times \text{Post}_t) + \varepsilon_{ipt}, \tag{1}$$

where Y_{ipt} is either an indicator for vaping in the past 30 days or an indicator for current smoking, Treated_p equals one for British Columbia and zero for control provinces, and Post_t equals one for survey years from 2020 onward. The interaction term $\text{Treated}_p \times \text{Post}_t$ captures the additional change in British Columbia relative to the change in the control provinces. The coefficient δ is the simple difference in differences estimate. The constant term α captures the mean outcome in the control provinces in the pre period. The coefficient β captures the pre period difference between British Columbia and the control provinces. The coefficient γ captures the common change from pre to post period that is shared by all provinces. In practice, equation (1) can be recovered either by regression or by comparing cell means across the four groups defined by province and period.

Our preferred specification introduces province and year fixed effects in order to control more flexibly for time invariant differences across provinces and common shocks in each year. For outcome Y_{ipt} we estimate

$$Y_{ipt} = \beta\,\text{tax_bc}_{pt} + \gamma_p + \lambda_t + u_{ipt}, \tag{2}$$

where γ_p are province fixed effects, λ_t are year fixed effects, and tax_bc_{pt} is an indicator that equals one for British Columbia in 2020 and later years. The error term u_{ipt} captures unobserved determinants of the outcome. The coefficient β measures the average effect of the tax on British Columbia relative to the control provinces after accounting for permanent differences across provinces through γ_p and common shocks in each year through λ_t. In this specification, identification comes from changes over time in British Columbia relative to changes over time in control provinces that did not introduce their own vaping taxes during the sample period.

To study dynamic adjustment and to assess the parallel trends assumption, we also estimate an

event study specification for vaping. Let D_{pt}^{τ} be an indicator for province p in year t that equals one if p is British Columbia and t is a particular year τ and zero otherwise. Taking 2019 as the base year, we estimate

$$Y_{ipt} = \gamma_p + \lambda_t + \sum_{\tau \neq 2019} \theta_\tau D_{pt}^{\tau} + u_{ipt}, \tag{3}$$

so that each coefficient θ_τ measures the difference between British Columbia and the control provinces in year τ relative to their difference in 2019. The fixed effects γ_p and λ_t again absorb province specific and year specific components. Under parallel trends, the pre tax coefficients θ_τ for years before 2019 should be close to zero. The post tax coefficients trace out how the gap between British Columbia and the control provinces evolves after the tax. Plotting θ_τ with confidence intervals provides a visual check of both pre trends and dynamic responses.

Before explaining the dynamic responses in detail, we use the simple two period difference in differences in equation (1) to summarise how average vaping and smoking changed in British Columbia and in the control provinces.

Table 1 summarises the mean vaping rate in British Columbia and in the control provinces before and after 2020. The control provinces had an average vaping rate of 0.0353 before 2020 and 0.0809 after 2020, so vaping increased by about 4.56 percentage points in the absence of the British Columbia tax. British Columbia started slightly above this level with a pre period rate of 0.0391 and reached 0.0794 after 2020, an increase of 4.03 percentage points. The difference in differences estimate is therefore

$$\delta_{\text{vape}} = (0.0794 - 0.0391) - (0.0809 - 0.0353) = -0.0053,$$

which corresponds to a 0.53 percentage point reduction in vaping in British Columbia relative to the control provinces. These numbers indicate that vaping continued to rise on average in both treated and control provinces, but that the increase was modestly smaller in British Columbia once the tax was in place.

Table 1: Simple difference in differences for vaping

	Pre < 2020	Post ≥ 2020	Difference (Post $-$ Pre)
Control provinces	0.0353	0.0809	0.0456
British Columbia	0.0391	0.0794	0.0403
Difference (BC $-$ control)	0.0038	-0.0015	-0.0053

Notes: Entries are mean vaping rates in the past 30 days. Control provinces exclude those that introduced their own provincial vaping taxes during the sample period. The bottom row reports differences between British Columbia and the control provinces.

Table 2 repeats the same exercise for current smoking. Here the control provinces show a decline from 0.1388 to 0.0967, a drop of 4.21 percentage points. British Columbia starts lower at 0.1102 and falls to 0.0744, a drop of 3.58 percentage points. The difference in differences estimate is

$$\delta_{\text{smoke}} = (0.0744 - 0.1102) - (0.0967 - 0.1388) = 0.0063,$$

which corresponds to a 0.63 percentage point increase in smoking in British Columbia relative to the controls. This simple comparison suggests that the tax period coincides with a slightly smaller decline in smoking in British Columbia than in the control provinces, although the change is small in absolute terms.

Table 2: Simple difference in differences for smoking

	Pre < 2020	Post ≥ 2020	Difference (Post $-$ Pre)
Control provinces	0.1388	0.0967	-0.0421
British Columbia	0.1102	0.0744	-0.0358
Difference (BC $-$ control)	-0.0286	-0.0223	0.0063

Notes: Entries are mean current smoking rates. The bottom row reports differences between British Columbia and the control provinces.

These simple comparisons suggest that the tax is associated with lower vaping but slightly higher smoking in British Columbia relative to the control provinces. We now turn to the fixed effects specification in equation (2), which uses within-province variation over time and controls for common year effects. Normally, when comparing two periods, these results would be the same as the simple difference in difference model. However, since we examine multiple time periods we uncover slightly different results.

Table 3 reports the main difference in differences estimates from the fixed effects models. Column (1) uses vaping in the past 30 days as the outcome for all ages combined. Once we control for permanent differences across provinces and for common year shocks, the tax indicator for British Columbia is slightly negative. This means that, on average, the tax is associated with a reduction in vaping prevalence of a little more than half a percentage point. Given the sampling variation in the data, this effect is not estimated very precisely. Therefore, we cannot rule out that the true effect is smaller or close to zero.

Columns (2) shows that for the youth ages 15-19, the tax is associated with a sizeable drop in vaping. The point estimate implies that the probability of vaping in the past 30 days falls by over one percentage points in British Columbia relative to the youth in the control provinces. This is a sizeable shift given the high baseline rates of vaping in this age group. Similarly to the overall population, this result lacks statistical significance. This age-specific result suggests that the tax is more effective on the youth. This is expected given that Young people are more likely to vape as seen in the preliminary analysis.

Column (3) turns to current smoking for all ages. Here the tax indicator is slightly positive, which means that smoking in British Columbia is higher after the tax than in the control provinces once we account for fixed effects. While the result lacks statistical precision, it shows a substitution effect between smoking and vaping.

Finally, column (4) uses an indicator for any nicotine use as the outcome. This variable equals one if a person either vapes or smokes. The estimate for the tax indicator in this column is extremely close to zero. This means that, once we combine vaping and smoking, the higher tax in British Columbia does not produce a detectable change in the share of people who use at least one nicotine product. Put differently, the tax appears to shift how some people consume nicotine, especially among young adults, but it does not change the overall fraction of the population that uses nicotine.

Table 3: Difference in differences estimates with province and year fixed effects

	Vaping in past 30 days		Smoking	Any nicotine
	All ages	15 to 19	All ages	All ages
tax_bc_{pt}	−0.0056	−0.0129	0.0036	0.0002
	(0.0035)	(0.0091)	(0.0036)	(0.0040)
Observations	66,478	14,360	66,478	66,478
R-squared	0.0129	0.0309	0.0062	0.0020
Province fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Notes: Each column reports coefficients from a separate linear probability model. The dependent variable is an indicator for vaping in columns 1 to 2, an indicator for current smoking in column 3, and an indicator for any nicotine use (vaping or smoking) in column 4. tax_bc_{pt} equals one for British Columbia in 2020 and later years and zero otherwise. Standard errors in parentheses are clustered at the province level. *** denotes significance at the 1 percent level.

The fixed effects estimates in Table 3 summarise the average impact of the tax across the whole post period. To see how this impact builds over time and to check the parallel trends assumption more directly, we next turn to the event study specification in equation (3). This model produces a set of year specific coefficients for British Columbia relative to the control provinces, normalised to zero in 2019. Figure 9 plots these estimates and their confidence intervals for vaping in the past 30 days.

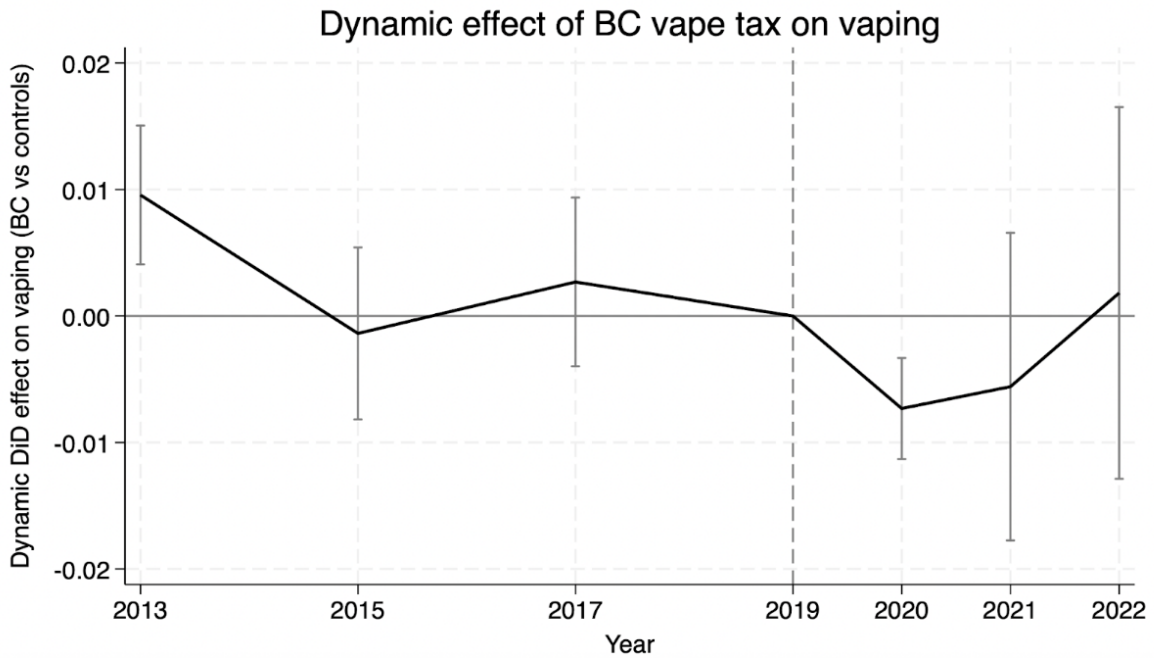

Figure 9: Event study estimates for vaping in BC relative to control provinces

Table 4: Event study regression for vaping with 2019 as base year

	Coefficient	Std. error
Treated province (BC)	-0.0004	(0.0030)
Year 2013	-0.0358^{***}	(0.0028)
Year 2015	-0.0247^{***}	(0.0035)
Year 2017	-0.0217^{***}	(0.0034)
Year 2020	0.0032	(0.0020)
Year 2021	0.0296^{***}	(0.0062)
Year 2022	0.0349^{***}	(0.0075)
BC $\times$ 2013	0.0096^{**}	(0.0028)
BC $\times$ 2015	-0.0014	(0.0035)
BC $\times$ 2017	0.0027	(0.0034)
BC $\times$ 2020	-0.0073^{**}	(0.0020)
BC $\times$ 2021	-0.0056	(0.0062)
BC $\times$ 2022	0.0018	(0.0075)
Constant	0.0584^{***}	(0.0030)
Province fixed effects	Yes	
Observations	66,478	
R^2	0.0129	
Clustering level	Province	

Notes: OLS regression of an indicator for vaping in the past 30 days on indicators for British Columbia, survey year, and their interactions, with province fixed effects. Year 2019 is the omitted base year. Standard errors in parentheses are clustered at the province level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 9 and Table 4 show that the estimated coefficients θ_τ are small in magnitude in the years before the tax. In 2013 the estimate is roughly $+0.01$, in 2015 it is around -0.001, and in 2017 it is about $+0.003$. All of these are close to zero, which suggests broadly similar pre tax trends between British Columbia and the control provinces, although the positive estimate for 2013 indicates a small initial level difference.

In the post tax period the picture changes. In 2020, the first year after the tax increase, the event study coefficient is approximately -0.0073 with an estimated standard error of about 0.0020. This

implies a reduction in vaping of around 0.7 percentage points in British Columbia relative to the control provinces, conditional on fixed effects, and it is statistically significant at roughly the 1 percent level. In 2021 the coefficient is about -0.0056, and in 2022 it is around $+0.0018$. These later estimates are smaller in magnitude and not statistically different from zero, and their confidence intervals are wide.

Taken together, the dynamic results show that the British Columbia tax operates as a short run shock rather than a permanent shift in behaviour. Vaping in the province falls relative to the control provinces in the first year after the tax, but this gap narrows over time and has largely closed by the end of the sample. By 2022 there is no clear evidence that the tax continues to hold vaping rates down, which suggests that users and markets adapt to the higher tax, for example by changing products, purchasing patterns or other aspects of nicotine use. On its own, a one time tax increase therefore does not appear to deliver a sustainable long run reduction in vaping participation at the population level. This has important policy implications. A sharp tax rise can slow or temporarily reverse the growth of vaping, especially among young adults. However, over time consumers adjust to the higher price. Maintaining the decrease in usage would likely require repeated adjustments to the tax.

An important event to note is the introduction of a federal excise tax on vaping products in 2022 that applies to all provinces at the same time. We did not have to account for this in the model because the impact of this national tax is absorbed by the year fixed effects in our models. In practice, this means that our difference in differences estimates capture how British Columbia evolves relative to other provinces after its own provincial tax, on top of any common change driven by the later federal tax.

The event study speaks to the credibility of the identification strategy. In the years before the tax, the estimated differences between British Columbia and the control provinces are small and do not move in a clear upward or downward pattern. This supports the parallel trends assumption that underlies the difference in differences design and strengthens the case for interpreting the post tax changes as a policy response rather than as a continuation of pre existing trends.

The empirical work faces several challenges that shape how these results should be interpreted. First, the tax change coincides with the onset of the COVID period, which affected social behaviour, incomes and daily routines across Canada. Year fixed effects help to absorb shocks that are common to all provinces, but any shocks that hit British Columbia differently from the control provinces at the same time as the tax may still bias the estimates. Second, the number of provinces is limited, so inference with province-level clustering rests on a relatively small number of clusters.

Third, the analysis combines data from two survey programs with different modes of data collection and slightly different questionnaires, so harmonisation may introduce measurement noise in the outcomes. Fourth, the sample becomes smaller once we split by age group, especially when focusing on a single treated province. This limits the precision of some subgroup estimates and may mask smaller effects. Finally, the public use data do not contain exact interview dates, so treatment is defined at the survey year level rather than at the level of months relative to the policy, which prevents a finer view of very short-run responses.

Nevertheless, we find that across all specifications, a consistent pattern emerges. The British Columbia vaping tax appears to reduce vaping, with strong and most precisely estimated effects among youth aged 15 to 19. Older adults also reduce vaping, but by a smaller amount. At the same time, there is an increase in smoking with no detectable change in overall nicotine use. The event study suggests that the main effect on vaping occurs in the first year after the tax and then fades. In the next section we discuss how these findings relate to public health and consumer welfare.

6 Conclusion

Policy makers in British Columbia chose to raise the provincial sales tax on vaping products from 7 percent to 20 percent in 2020, with the stated aim of reducing youth vaping. The purpose of this paper was to analyze whether this policy decreased vaping and if it had the unintended consequence of increasing smoking prevalence.

We used a difference in differences framework to compare British Columbia with other provinces before and after the tax. Our first objective was to measure the impact of the tax on vaping itself. For the population as a whole, we found a smaller reduction in vaping of 0.56 percentage points. Allowing the effect to vary by age, we found that for people aged fifteen to nineteen, the tax was associated with a larger fall in vaping of 1.3 percentage points. Taken together, these results suggest that the tax did reduce vaping, with the strongest response among youth, who are the group most heavily involved in vaping at baseline.

Our second objective was to see whether higher vaping taxes led to substitution toward smoking or changed total nicotine use. In the fixed effects models, the vaping tax led to an increase in smoking of 0.36 percentage points. These results suggest that the British Columbia tax reduced vaping for some age groups while causing an offsetting rise in smoking. The event study analysis adds a time dimension. It shows a distinct drop in vaping in the first year after the tax, followed by smaller and less precise differences in later years. This points to a short-run response to the tax rather than a large and permanent shift in behaviour.

These findings add a Canadian perspective to an international literature that has mostly focused on the United States. Our results are consistent with the idea that vaping and smoking are substitutes for many users. From a policy point of view, this suggests that a provincial vaping tax can decrease vaping, but at the cost of increasing smoking rates.

Our analysis has several limitations that should be considered when interpreting the results. All of our outcomes are based on self-reported survey data, so vaping and smoking may be underreported, especially among younger respondents. The public use files do not include exact interview dates, preventing analysis of precise responses around the announcement and implementation of the tax. Our design relies on a single treated province but a limited number of control provinces. Additionally, we do not observe quantities or prices, so we cannot estimate full demand elasticities or welfare effects. Finally, we narrow our heterogeneity analysis to age groups and do not study differences by income, gender, baseline smoking status, or intensity of use, all of which could matter for the impact of the tax.

These limitations point to several directions for future work. One natural extension would be to combine survey data with retail scanner data or administrative tax records. This enables the analysis of how the taxes pass through to retail prices. Another is to follow Canadian provinces over a longer horizon as more vaping and nicotine regulations are introduced to study how overlapping provincial and federal policies interact. Future work could also explore richer forms of heterogeneity, for example by separating effects on initiation and quitting, or by comparing heavy and light users. Welfare-oriented work could combine behavioural responses with models of health risk to quantify whether a given mix of cigarette and vaping taxes improves or worsens public health. Our study cannot say how the tax changed the intensity of use among continuing users, how it affected brand choice, flavours or informal markets, or whether some groups bore larger financial and health costs than others. We also do not take a stand on the optimal level of vaping taxation or on the best combination of cigarette and vaping taxes. These remain open questions, and answering them will require data that link individual behaviour to prices, products and health outcomes in a more detailed way than is possible here.

7 Appendix

A Additional figures

A.1 Age patterns in British Columbia

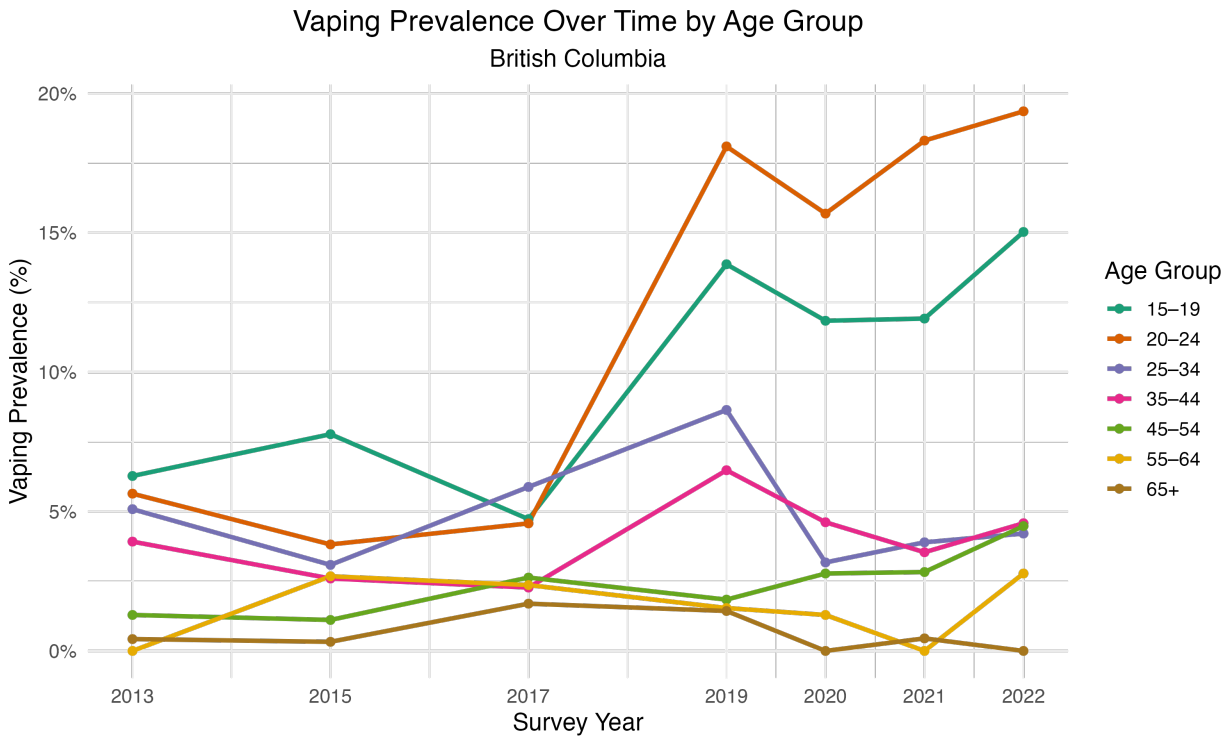

Figure 10: Vaping prevalence by age group in British Columbia, 2013–2022.

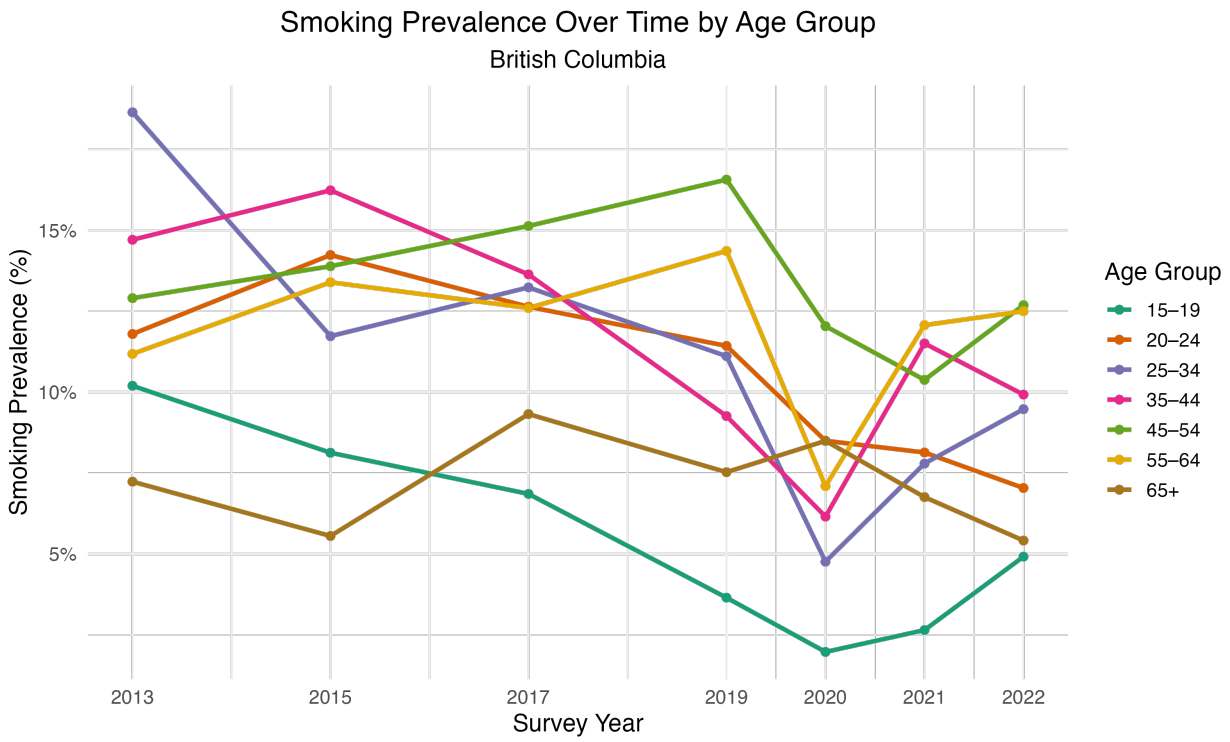

Figure 11: Smoking prevalence by age group in British Columbia, 2013–2022.

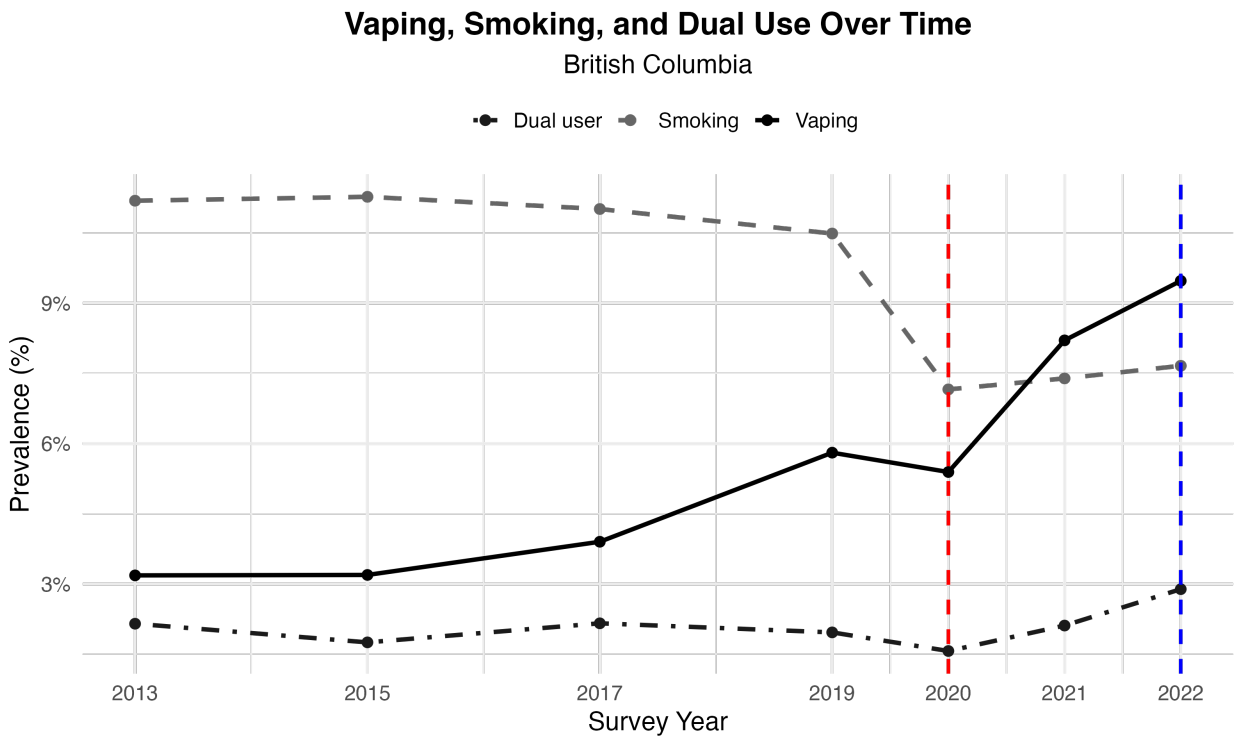

Figure 12: Vaping only, smoking only and dual use in British Columbia, 2013–2022.

A.2 Provincial comparisons

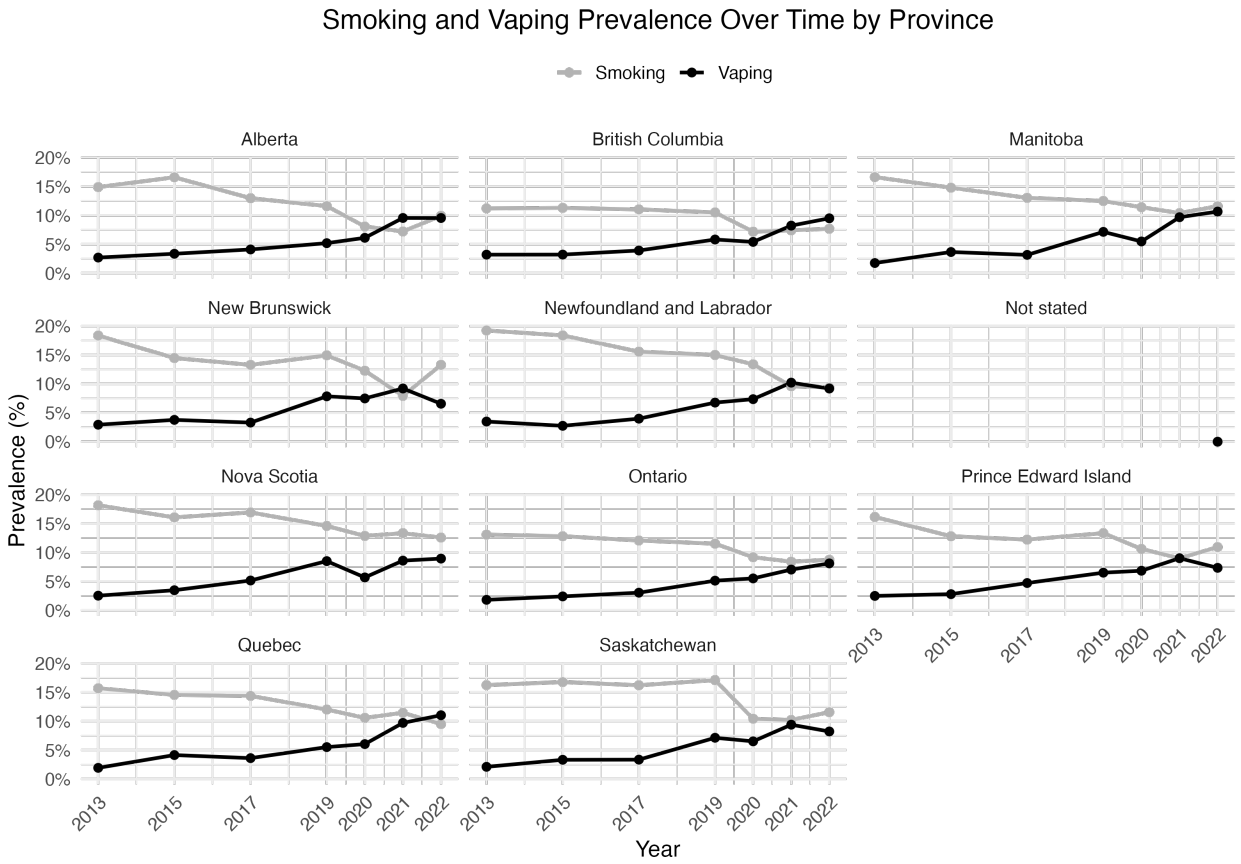

Figure 13: Smoking and vaping prevalence by province over time.

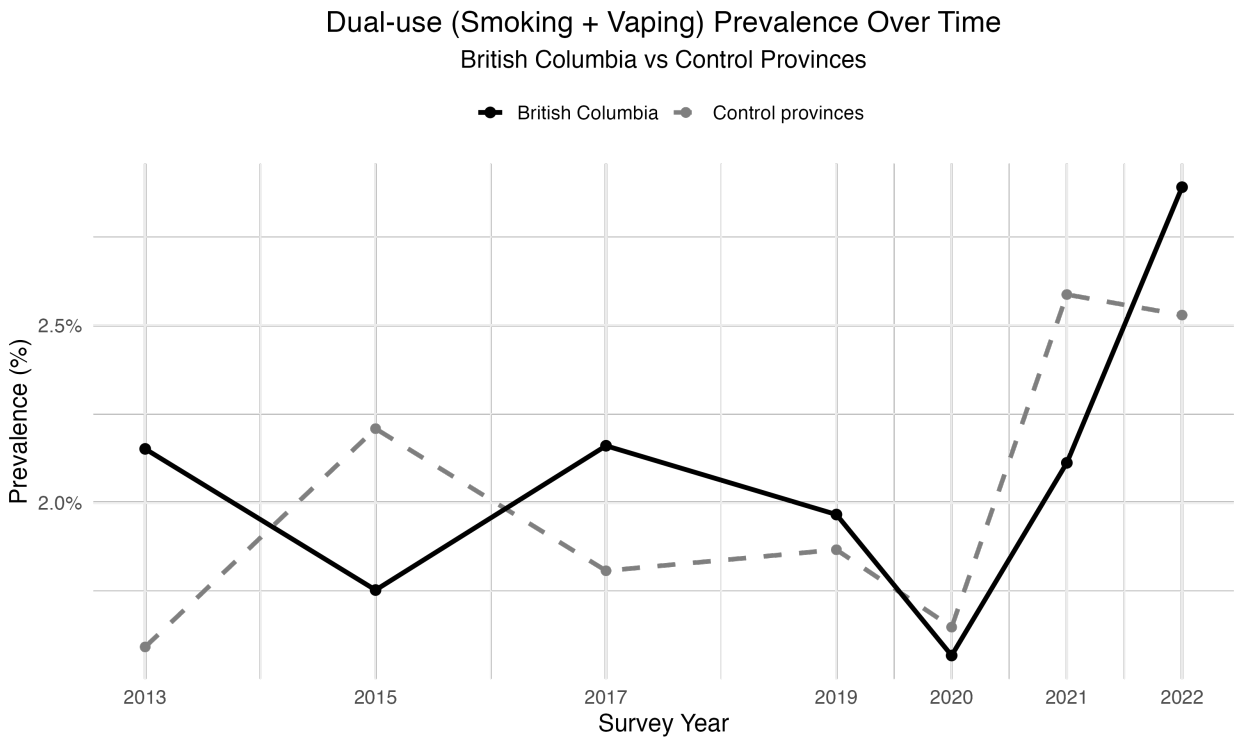

Figure 14: Dual use in British Columbia and control provinces, 2013–2022.

A.3 Difference in differences estimates

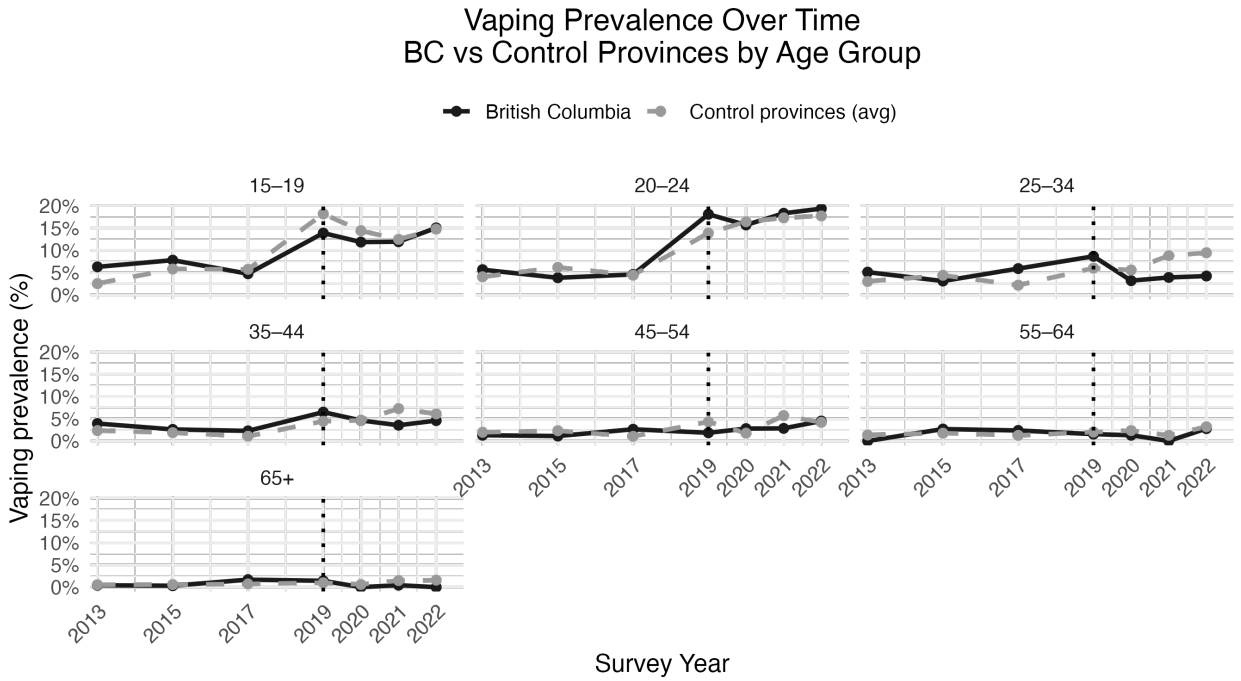

Figure 15: Difference in differences estimates for vaping in British Columbia relative to control provinces.

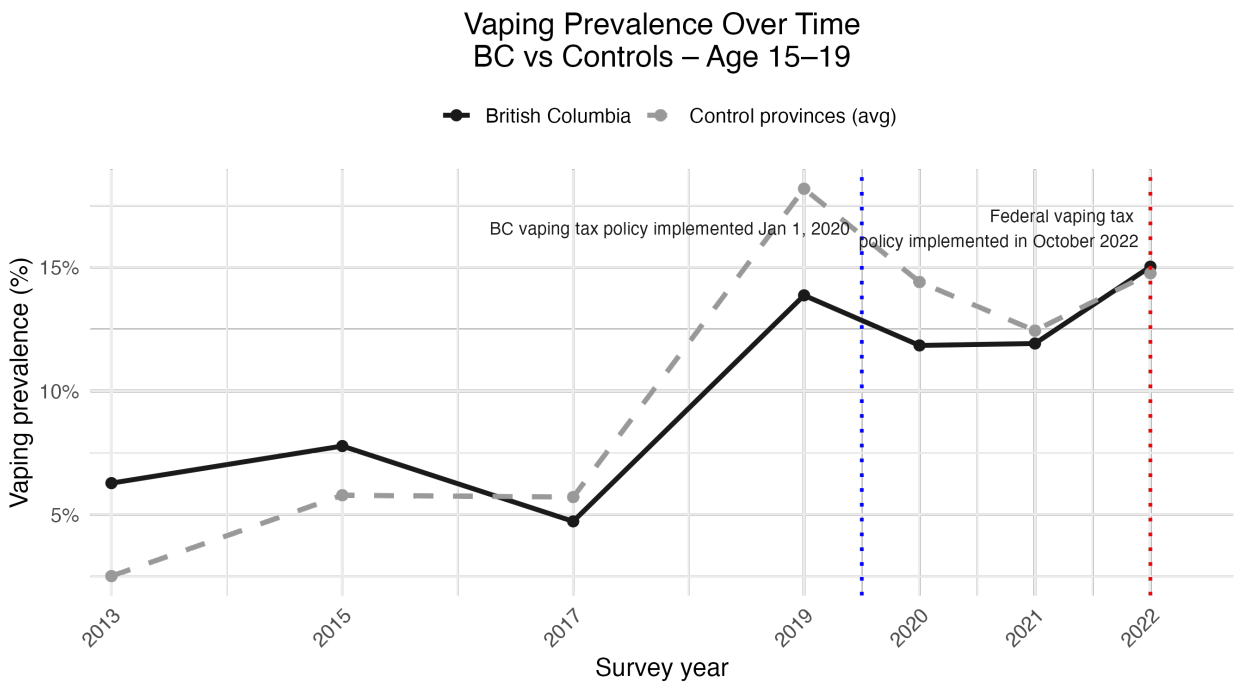

Figure 16: Difference in differences estimates for vaping among youth aged 15–19 in British Columbia and control provinces.

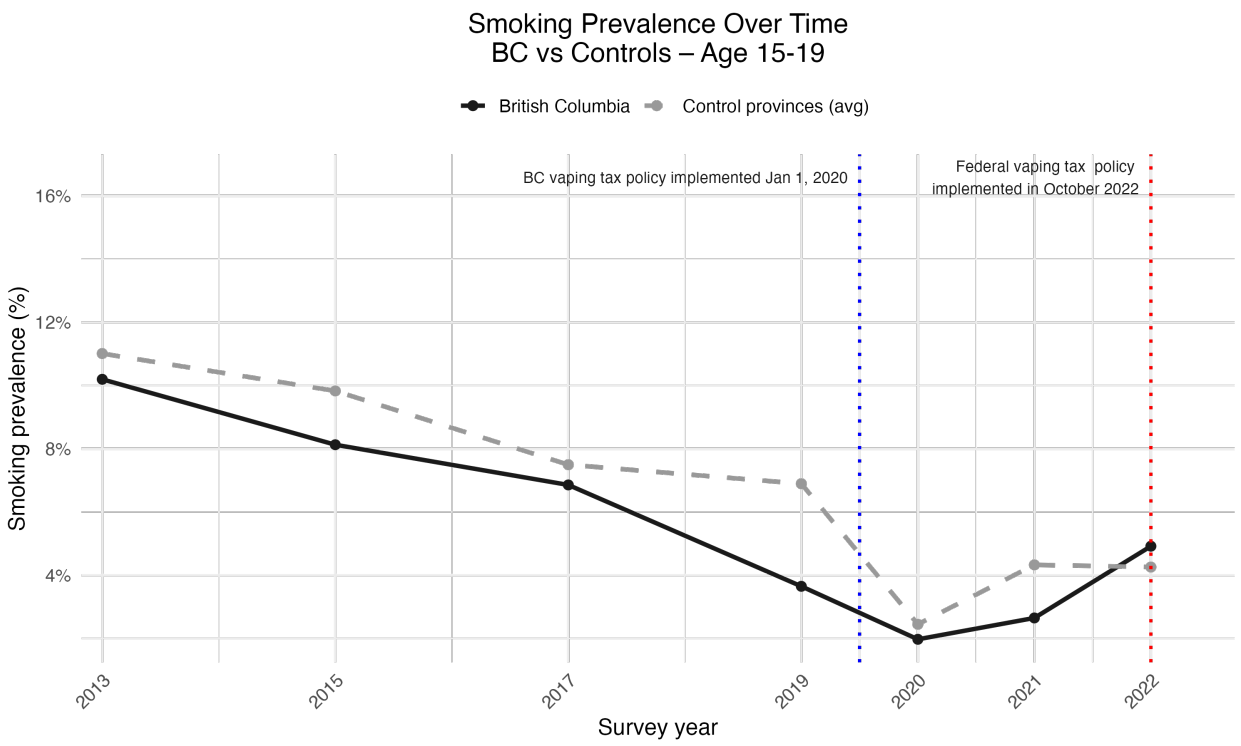

Figure 17: Difference in differences estimates for smoking among youth aged 15–19 in British Columbia and control provinces.

REFERENCES

Abadie, A. (2021). Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, 59(2):391–425.

Abouk, R., Adams, S., Feng, B., Maclean, J. C., and Pesko, M. F. (2023). Intended and unintended effects of e-cigarette taxes on youth tobacco use. *Journal of Health Economics*, 87:102720.

Brill, A., Satel, S., and Viard, A. D. (2014). Should e-cigarettes be taxed? *Tax Notes*, 143(2). Tax Policy Center and American Enterprise Institute report.

Friedman, A. S. and Pesko, M. F. (2022). Young adult responses to taxes on cigarettes and electronic nicotine delivery systems. *Addiction*, 117(12):3121–3128.

Hammond, D., Reid, J. L., Burkhalter, R., and Rynard, V. L. (2020). E-cigarette marketing regulations and youth vaping: Cross-sectional surveys, 2017–2019. *Pediatrics*, 146(1):e20194020.

Irvine, I. and Hampsher-Monk, S. (2022). Early indications of consumer responses to canada's nicotine limits on e-cigarettes. BOTEC Analysis report for the Foundation for a Smoke-Free World. April 2022.

Pesko, M. F., Courtemanche, C. J., and Maclean, J. C. (2020). The effects of traditional cigarette and e-cigarette taxes on adult tobacco product use. *Journal of Risk and Uncertainty*, 60(3):229–258.

Saffer, H., Dench, D., Grossman, M., and Dave, D. (2020). E-cigarettes and adult smoking: Evidence from minnesota. *Journal of Risk and Uncertainty*, 60(3):207–228.

Statistics Canada (2016). Canadian tobacco, alcohol and drugs survey microdata file, 2015. Catalogue no. 82M0020X2016001, Canadian Tobacco, Alcohol and Drugs Survey (CTADS) public use microdata. Data for CTADS survey cycles 2013, 2015 and 2017, conducted by Statistics Canada on behalf of Health Canada.

Statistics Canada (2023). Canadian tobacco and nicotine survey: Public use microdata file. Catalogue no. 13-25-0001, Canadian Tobacco and Nicotine Survey (CTNS) public use microdata. Data for survey cycles 2019, 2020, 2021 and 2022, conducted by Statistics Canada on behalf of Health Canada.